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SUPERSEDING
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MILITARY SPECIFICATION

PUMPING ASSEMBLIES, DEEP-WELL, GASOLINE-ENGINE DRIVEN: HELICAL-ROTOR AND TURBINE TYPES

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers gasoline-engine driven, water-lubricated, turbine and helical-rotor, deep-well pumps complete with all necessary accessories and connections.

1.2 Classification. Pumping assemblies shall be of the following classes and sizes, as specified (see 6.2):

Class A - Turbine
  Size 1 - 200 gallons per minute (gpm) at 200-foot total head.
  Size 2 - 200 gpm at 350-foot total head.
  Size 3 - 50 gpm at 250-foot total head.
Class B - Helical Rotor - 50 gpm at 250-foot total head.

FSC 4320

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, VA 22060, by using the self-addressing Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

FF-B-171	-	Bearings, Ball, Annular (General Purpose).
WW-P-521	-	Pipe Fittings, Flange Fittings, and Flanges: Steel and Malleable Iron (Threaded and Butt-Welding) Class 150.
WW-V-58	-	Valves, Gate, Cast-Iron, Threaded and Flanged (for Land Use).
XX-P-746	-	Pump, Inflating, Manual, (for Rubber Tires).

MILITARY

MIL-V-173	-	Varnish, Moisture-and-Fungus-Resistant (for treatment of Communications, Electronic, and Associated Equipment).
MIL-P-514	-	Plate, Identification, Instruction and Marking, Blank.
MIL-T-704	-	Treatment and Painting of Materiel.
MIL-P-710	-	Pump, Deep-Well, Packaging of.
MIL-F-3541	-	Fittings, Lubrication
MIL-G-3859	-	Grease Guns, Hand-Operated, Lever, Push and Screw Type.
MIL-G-10924	-	Grease, Automotive and Artillery.
MIL-G-23827	-	Grease, Aircraft and Instrument, Gear and Actuator Screw.
MIL-E-52649	-	Engine Cold Starting Aids, Ether Fuel Primers.

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STANDARDS

FEDERAL

FED-STD-H28/7	-	Screw-Thread Standards For Federal Services, Section 7 American Standard Pipe Threads (except Dryseal and Hose Coupling Types).

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-130	-	Identification Marking of US Military Property.
MIL-STD-461	-	Electromagnetic Interference Characteristics Requirements for Equipment.
MIL-STD-1188	-	Commercial Packaging of Supplies and Equipment.
MIL-STD-1410	-	Methods for Selection of Industrial Engines for End Item Application.
MIL-STD-1472	-	Human Engineering Design Criteria for Military Systems, Equipment and Facilities.
MIL-STD-1474	-	Noise Limits for Army Materiel.
MS35647	-	Padlock, Key Operated.
MS35829	-	Hinge, Butt, Continuous (Piano) .075 Thickness, .187 and .250 Pin Dia.
MS35844	-	Plug, Machine Thread, Magnetic: Drain.
MS35932	-	Cock, Plug: Three Way, Threaded Type, 125 PSI.
MS49005	-	Plugs, Pipe-Headless, Aluminum, Iron, or Steel.

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MS49006 - Plugs, Pipe, Magnetic-Headless; Aluminum, Iron or Steel.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

B40.1 - Gages-Pressure and Vacuum Indicating Dial Type-Elastic Element.

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A 36 - Structural Steel.
A 48 - Gray Iron Castings.
A 108 - Steel Bars, Carbon, Cold-Finish Standard Quality.
A 120 - Pipe Steel, Black and Hot-Dipped Zinc-Coated (Galvanized) Welded and Seamless, for Ordinary Uses.
A 269 - Seamless and Welded Austenitic Stainless Steel Tubing for General Service.
A 314 - Stainless and Heat-Resisting Steel Billets and Bars for Forging.
A 582 - Free-Machining Stainless and Heat Resisting Steel Bars, Hot-Rolled or Cold Finished.
B 62 - Composition Bronze or Ounce Metal Castings.
B 165 - Nickel-Copper Alloy (UNS N04400) Seamless Pipe and Tube.

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B 584 - Copper Alloy Sand Castings for General Applications.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

Boiler and Pressure Vessel Code, Section IX, Welding Qualifications.

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017.)

AMERICAN WATER WORKS ASSOCIATION (AWWA)

E101 - American National Standard for Vertical Turbine Pumps - Line Shaft and Submersible Types.

(Application for copies should be addressed to the American Water Works Association, 6666 West Quincy Avenue, Denver, CO 80235.)

AMERICAN WELDING SOCIETY (AWS)

D1.1 - Structural Welding Code, Section 5, Qualification.

(Application for copies should be addressed to the American Welding Society, 2501 Northwest Seventh Street, Miami, FL 33125.)

HYDRAULIC INSTITUTE (HI)

Standards of the Hydraulic Institute.

(Application for copies should be addressed to the Hydraulic Institute, 712 Lakewood Center North, Cleveland, OH 44107.)

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

J429 - Mechanical and Material Requirements for Externally Threaded Fasteners.

(Application for copies should be addressed to the Society of Automotive Engineers, 400 Commonwealth Drive, Warrandale, PA 15096.)

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SOUTHERN CALIFORNIA AIR POLLUTION CONTROL DISTRICT

102 - Photochemically Reactive Solvents. Rule 102.

(Application for copies should be addressed to the Southern California Air Pollution Control District, 434 So. San Pedro Street, Los Angeles, CA 90013.)

STATE OF CALIFORNIA DEPARTMENT OF MOTOR VEHICLES

California Vehicle Code.

(Application for copies should be addressed to the State of California, Department of Motor Vehicles, 2570 24th Street, Sacramento, CA 95809.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Description. The pumping assemblies shall consist of a power-and-drive unit mounted on a skid base, a column pipe and shaft assembly with the pumping element, and the necessary connections and accessories as specified herein.

3.1.1 Major components. The power and drive unit shall consist of a gasoline engine, a power take-off (clutch), a drive shaft, a discharge head, and an angle drive, all mounted on a skid base. The pumping element and the column pipe and shaft assembly shall be suspended vertically from the discharge head. The pumping assemblies shall be furnished with the quantities of column shafting, column pipe, and air line as specified in Table I.

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TABLE I. Column shafting, column pipe, and air line.

Pumping assembly classification	Length (feet)		
	Column shafting	Column pipe	Air line
Class A Turbine			
Size 1	150	150	155
Size 2	150	150	155
Size 3	200	200	205
Class B - Helical rotor	200	200	205

3.2 First article. Unless otherwise specified (see 6.2) the contractor shall furnish one or more pumping assemblies for examination and test within the time frame specified (see 6.2), to prove prior to starting production, that his production methods will produce pumping assemblies that comply with the requirements of this specification. Examination and tests shall be as specified in Section 4 and unless otherwise specified herein, all examination and tests shall be conducted by the contractor subject to surveillance and approval by the Government (see 6.3). When specified (see 6.2), the Government will conduct any or all of the first article examination and tests as specified (see 6.2).

3.3 Initial production. When specified (see 6.2), the contractor shall furnish to the Government one or more pumping assemblies for inspection as specified in 4.4.

3.4 Material. Material shall be as specified herein. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification (see 6.8).

3.5 Threads. Threads shall conform to FED-STD H28/7.

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3.6 Environmental Conditions.

3.6.1 Operating temperature. The pumping assemblies shall perform as specified herein in any ambient temperature from plus 120° F (plus 49° C) to minus 25° F (minus 32° C).

3.6.2 Storage temperature. The pumping assemblies shall not be damaged by storage in any ambient temperature from plus 155° F (plus 68° C) to minus 65° F (minus 54° C).

3.7 Safety. All rotating parts and parts subject to high operating temperatures that are of such nature or so located as to become a hazard to operating personnel shall be insulated, fully enclosed, or guarded. Protective devices shall not impair the operation functions.

3.8 Human factors engineering. The pumping assemblies shall conform to the applicable human factors engineering criteria as specified in MIL-STD-1472.

3.8.1 Noise limits. The noise of the pumping assemblies (excluding horns and warning alarms) shall not exceed the limits specified in MIL-STD-1474, Category D for steady-state noise when tested in accordance with 4.6.2.7.1. The sound level shall comply with the exterior noise requirements of MIL-STD-1474 when tested in accordance with 4.6.2.7.2 (aural nondetectability and impulse noise do not apply). The precautions of MIL-STD-1474 (see 6.6) shall be provided if the sound level exceeds 85 dB(A) when tested in accordance with 4.6.2.7.3. Hearing protectors shall not be supplied with the pumping assemblies. Hazard signs, when required, shall state "CAUTION, HEARING PROTECTION REQUIRED". Plates shall be in accordance with MIL-P-514, Type III, Composition C of Type I, Grade A, Class 1 material.

3.9 Reliability. The specified mean-time-between-failure (MTBF) of the pumping assemblies shall be 330 hours when tested in accordance with 4.6.2.8.

3.10 Maintainability. Any maintenance operation performed on the pumping assemblies during testing in accordance with 4.6.2.9 shall be accomplished by one man at organizational level using common maintenance tools normally available or special tools furnished with the pumping assembly. None of the major components of the pumping assembly shall

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require replacement, rebuild, or overhaul during the test. The ratio of active maintenance manhours (including both scheduled and unscheduled maintenance) expended during the test to the total unit operating time shall not exceed 0.10. A maintenance schedule shall be furnished prior to the start of test.

3.11 Ease of maintenance. All major assemblies and installed attachments shall be accessible for maintenance, repair, and replacement without the removal of other major assemblies and installed attachments not normally removed. Covers or plates which must be removed for component adjustment, repair, replacement, or maintenance shall be equipped with quick-disconnect fasteners. Drain outlets shall be located for accessibility.

3.12 Gasoline engine. For the purpose of selecting a 4-stroke cycle engine, the pumping assemblies shall be classified as a Class II end item in accordance with MIL-STD-1410. The engine, including all systems, components, and accessories, shall conform to MIL-STD-1410, Class II, except as otherwise specified herein. The engine shall have both manual and electrical starting systems, and engine housing; and shall be liquid or air-cooled, as specified (see 6.2). The engine shall be furnished with a 12-volt cranking and battery-charging system. The battery-charging system shall be the alternator type of not less than 35-ampere capacity.

3.12.1 Engine operation requirements. The engine operational requirements and maximum engine operating limits shall be in accordance with MIL-STD-1410 under any of the requirements and conditions specified herein. The engine, as applied to the pumping assemblies shall meet the applicable State of California exhaust emissions standards.

3.12.2 Governing system. The engine shall be provided with a variable-speed governor, gear-driven from the engine. Provisions shall be made for trimming governed speed setting throughout the entire load range while the engine-driven equipment is in operation. The engine speed at any load between no-load and full rated load shall be not less than rated load speed nor exceed rated load speed by more than 10 percent as specified in MIL-STD-1410.

3.12.3 Control-device system. The liquid-cooled engine shall be provided with a high-cooling-liquid-temperature device for stopping the engine when the engine reaches an overheated condition. The air-cooled engine shall be provided with a high-cylinder-head-temperature device for stopping the engine when the engine reaches an overheated condition.

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Both liquid- and air-cooled engines shall be provided with a low-lubricating-oil-pressure control device for stopping the engine when a low-lubricating-oil-pressure condition exists. Each of the safety devices shall have individual signals to indicate which deficiency has caused the engine shutdown.

3.12.4 Manual engine controls. Manual engine controls shall include an ignition switch, a choke, an override switch, and a throttle. The override switch shall be the spring-loaded automatic return type and shall activate the engine ignition system by bypassing the low oil pressure control device to facilitate starting when the oil pressure is low. When the override switch is released, it shall automatically return to a position that will provide low oil pressure protection for the engine. The throttle shall be the "T" handle positive locking type and shall permit operation of the engine at any selected speed from idling to maximum governed speed.

3.12.5 Fuel system. The fuel system shall be provided with a 3-way selector valve conforming to MS35932-1. The auxiliary connection of the selector valve shall be provided with a plug conforming to MS49005-2. A corrosion-resistant circular faceplate shall be mounted on the fuel selector valve and shall be marked "OFF", "UNIT TANK" and "AUXILIARY". A fuel tank having a capacity for not less than 4 hours of continuous, rated-load operation shall be provided. The fuel tank shall be mounted in such a manner as to prevent fuel from spilling on the engine during refueling.

3.12.6 Instrumentation. An instrument panel shall be furnished and shall be provided with the following instruments:

(a) Gage, pressure, lubricating oil.
(b) Indicator, battery-generator.
(c) Indicator, cooling liquid temperature (when liquid-cooled engine is provided).
(d) Tachometer.

3.12.7 Cold starting aid. When specified (see 6.2), an engine starting aid conforming to MIL-E-52649 shall be furnished, type and size required to start the engine at minus 25° F (minus 32° C).

3.12.8 Air cleaners. The engine air cleaners shall be the heavy duty dry type.

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3.12.9 Oil filters. The engine shall be equipped with full flow type oil filters.

3.13 Column pipe and shaft assembly.

3.13.1 Column pipe. The column pipe shall be black steel pipe conforming to ASTM A 120, sizes as specified herein. The pipe shall be furnished in nominal 10-foot lengths, except for one 5-foot length adjacent to the discharge casting and one 5-foot length adjacent to the pumping element. Ends of pipe shall be machined to butt and shall be threaded and furnished with couplings. The couplings shall be steel conforming to ASTM A 120, sizes as specified herein. The column pipe and couplings shall have straight threads which will make up into a butt type joint. The coupling shall be internally threaded American Standard Straight pipe threads in pipe coupling (NPSC). The pipe shall have American Standard straight pipe threads (NPS) to match with the coupling.

3.13.2 Column shafting. The column shafting shall be 1 inch in diameter and shall be steel conforming to ASTM A 108, UNS G10450; except that the top shaft section shall be steel conforming to ASTM A 314 UNS S41600. The shafting shall be furnished in 10-foot lengths, except for one 5-foot length adjacent to the pumping element and one length not less than 5 feet nor more than 10 feet adjacent to the discharge casting. The shafting shall be turned, ground, and polished, and the ends of each shaft length shall be machined to butt together. The shaft lengths shall be threaded at the ends and shall be provided with couplings. The couplings shall be steel conforming to ASTM A108, UNS G10450. Coupling and shaft threads shall be such that operation of the pumping assembly will not cause the shaft and coupling to become uncoupled. Abrasion-resistant, water-lubricated, rubber or graphite-impregnated bronze shaft bearings, supported in bronze spiders, shall be provided at each pipe joint, and shall be spaced to insure satisfactory operation at 3,600 revolutions per minute (rpm) for Class A turbine pumping assemblies, and 1,800 rpm for Class B helical rotor pumping assemblies. The shafting shall be fitted with sleeves of either corrosion-resistant steel conforming to ASTM A 269, Type TP304, or nickel-copper alloy conforming to ASTM B 165 at the bearing points. Bearing surfaces of the shaft sleeve shall be machined to a finish of 32 micro-inches. The shafting shall be adjustable by means of a top shaft nut.

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3.14 Accessories. The pumping assemblies shall be provided with the following accessories.

3.14.1 Discharge valve. Discharge valves shall conform to WW-V-58, Type I, Class 1, designation non-shock (NS); Sizes as specified in 3.23 and 3.24.

3.14.2 Discharge pressure gage. The discharge pressure gage shall conform to ANSI B40.1, Grade A, and shall have a range from zero to 200 pounds per square inch (psi). The gage shall have a clear-plastic dial-face lens.

3.14.3 Combination strainer and foot valve. A combination one-piece bronze strainer and rubber-seated foot valve shall be provided on the suction inlet of the pumping element. The strainer openings shall pass particles no greater than 75 percent of the cross-sectional dimension of the smallest water passages of the entire system. The net water passage area of the strainer shall be not less than two times the cross-sectional area of the column pipe. The valve shall withstand a hydrostatic pressure of not less than 150 psi without leakage, failure, or permanent deformation.

3.14.4 Handling attachments. The discharge head shall be furnished with handling attachments capable of withstanding 2.5 times the combined weight of the pump, column piping, column shafting, discharge head, water contained in the column piping, and the angle drive. The handling attachments shall be located to permit lifting the discharge head and attachments independently of the skid base and other drive components.

3.14.5 Water-level indicator. A water-level indicator consisting of the following shall be provided:

(a) Hand pump conforming to XX-P-746, Class A.
(b) Air valve.
(c) Air pressure gage graduated in feet of water with minimum dial range of from zero to 231 feet.
(d) Air line fabricated from 1/4-inch, standard weight, zinc-coated iron pipe, conforming to ASTM A 120 and furnished with the necessary zinc-coated, malleable-iron pipe fittings conforming to WW-P-521, Type II. The length of the air line shall be as specified in Table I.

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3.15 Power take-off unit. A power take-off unit shall be provided and shall consist of a clutch, housing, shaft, bearings, and all accessories necessary to transmit maximum engine brake horsepower to the drive shaft.

3.16 Alinement. The alinement of the engine and the pump angle drive shall fall within the misalinement tolerance of the drive shaft assembly.

3.17 Drive shaft assembly. The drive shaft assembly shall transmit power from the power take-off to the pump angle drive through an arrangement that will not introduce torsional vibration. Couplings used in the drive shaft assembly shall be the dry, flexible type and shall permit angular and linear misalinement without adverse effect. The drive shaft assembly shall be capable of transmitting the maximum engine horsepower without failure of permanent deformation.

3.18 Pump angle drive. The pump angle drive shall be a right angle bevel-gear drive. Anti-friction bearings shall be used throughout. The bearings shall conform to FF-B-171, Type and Class optional. The angle drive shall permit vertical adjustment of the pump shaft. The angle drive cover and casing shall be gray iron castings conforming to ASTM A 48, Class 30, and shall be bolted to the discharge casting.

3.19 Pump discharge head. A discharge head shall be provided and shall be a gray iron casting conforming to ASTM A 48, Class 30. The discharge head shall be provided with a packing box or it may be cast integrally. The packing-box assembly, as a minimum, shall consist of a cast iron packing box conforming to ASTM A 48, Class 30; bronze packing gland conforming to ASTM B 62, bronze bearing bushing, conforming to ASTM B 584; and molded graphited asbestos packing. A rubber slinger shall be provided on the shaft above the packing gland. The packing assembly shall minimize leakage between the column shafting and discharge head casting. The discharge head shall be bolted to the pump angle drive.

3.20 Skid base. The skid base shall be structural steel conforming to ASTM A 36. The skid base shall maintain alinement when the pumping assembly is being towed and handled in the field. The skids shall be provided with upturned ends and shall include an attachment for towing at each end of the skid base. The towing attachment shall be located so that the possibility of over-turning the pumping assembly is minimized and positive tracking is assured when the unit is being towed.

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3.21 Toolbox. A steel toolbox not less than 0.0747-inch (14 gauge) in nominal thickness shall be attached to the assembly. The toolbox shall contain all the tools required for the general maintenance and operation of the pumping assembly. The toolbox lid shall be secured to one edge of a side by a continuous hinge conforming to MS35829 extending the full length of the box edge. The lid shall be held in the closed position by not less than two tension-type fasteners with strikers. A hasp shall be located to secure the lid to the box and to accomodate a lock conforming to MS35647. The toolbox shall be mounted in an accessible, protected location permitting the lid to be open through an arc of 90 degrees. The lid shall be held fully open by the weight of the lid or a suitable fastener. Adequate drainage of the toolbox shall be provided as mounted.

3.22 Lifting attachment. The power and drive unit shall be provided with lifting attachments. The attachments shall enable the power and drive unit to be lifted in its normal operating position. The attachments shall be located so that clearance is maintained between the slings and all parts of the power and drive unit being lifted. Spreader bars are permitted. Each attachment shall carry not less than 2.5 times its share of the total load imposed on the attachments. Attachments for multiple slings shall be designed so that the attached slings will converge not more than 24 feet above the lowest extremity of the load when such limitation is possible. Attachments shall be positioned so that the lifting strain will be in line with the longitudinal axis of the eye of the attachment. The inside diameter of the lifting eye shall be not less than 3 inches. Structural members to which the attachments are fastened and fastenings shall withstand all stresses imposed by the attachments. Where required, reinforcements shall be used to meet the requirements specified herein.

3.23 Class A turbine. The Class A turbine pump shall be furnished with column pipe as specified in 3.13.1 and column shafting as specified in 3.13.2. The discharge head assembly shall not leak or show permanent deformation when subjected to a hydrostatic pressure of not less than 1.5 times the shutoff pressure. The discharge head assembly shall be furnish with a gate valve as specified in 3.14.1.

3.23.1 Size 1 pump. The Size 1 pump shall be furnished with column pipe and column shafting of the length shown in Table I. The column pipe shall be 4-inch diameter and suitable for use in a 6-inch inside diameter (ID) well casing. The discharge head gate valve shall be 4-inch nominal size.

3.23.2 Size 2 pump. The Size 2 pump shall be furnished with column pipe and column shafting of the length shown in Table I. The column pipe shall be 4-inch diameter and suitable for use in 6-inch ID well casing. The discharge head gate valve shall be 4-inch nominal size.

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3.23.3 Size 3 pump. The Size 3 pump shall be furnished with column pipe and column shafting of the length shown in Table I. The column pipe shall be 2.5 inch diameter and suitable for use in a 4-inch ID well casing. The discharge head gate valve shall be 2-inch nominal size.

3.23.4 Pumping element. The pumping element shall be water-lubricated vertical turbine, centrifugal type and shall consist of a series of individual turbine bowls screwed or bolted together. The bowls shall be gray iron castings conforming to ASTM A 48, Class 30A or Class 35A, and shall contain impellers on a common shaft of polished corrosion-resistant steel conforming to ASTM A 582, UNS S41600. The pumping assembly shall not leak (except around shaft packing) or show permanent deformation when subjected to a hydrostatic pressure of not less than 1.5 times the shutoff pressure.

3.23.4.1 Impellers. The impellers shall be bronze conforming to ASTM B 62. Impellers shall be cast in one piece and machined. All impellers shall be fixed rigidly and concentrically to the impeller shaft in such a manner that the impellers can be removed. Semi-enclosed impellers shall be so designed that after wear occurs, capacity and head can be restored by replacing the impellers without installing new pump bowls. Abrasion-resistant, water-lubricated, rubber or bronze bearings shall be provided on both sides of each impeller. When the bearings are bronze, the bronze shall conform to ASTM B 584 UNS C93800.

3.23.4.1.1 Impeller balance. The impellers shall be statically balanced prior to assembly. Tolerance for the static balance shall be a maximum of 0.50-inch-ounce plus 0.25-inch-ounce per inch of impeller diameter. The resonant frequencies of the assembled pumping element shall occur at speeds above the maximum operating speed of the pumping assembly.

3.23.5 Head capacity. The Class A pumping assembly shall deliver the following capacity when operated at speeds not exceeding 3,600 rpm.

3.23.5.1 Size 1. The Size 1 pumping assembly shall deliver 200 gpm at 200 feet total head.

3.23.5.2 Size 2. The Size 2 pumping assembly shall deliver 200 gpm at 350 feet total head.

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3.23.5.3 Size 3. The Size 3 pumping assembly shall deliver 50 gpm at 250 feet total head.

3.24 Class B, helical rotor. The Class B pumping assembly shall be furnished with 2-1/2 inch diameter column pipe as specified in 3.13.1 and column shafting as specified in 3.13.2. Lengths of column pipe and column shafting shall be as specified in Table I. The column pipe couplings shall be such that they can be used inside a 4-inch ID well casing. Stabilizers and rubber retainers shall be furnished for use in both 4-inch ID and 6-inch well casings. The discharge head assembly shall not leak or show permanent deformation when subjected to a hydrostatic pressure of not less than 200 psi. The discharge valve shall be a 2-inch gate valve as specified in 3.14.1. A pressure relief valve which can be set to release at 35 psi, 60 psi, and 100 psi, shall be provided. The relief valve shall be set to release at 100 psi.

3.24.1 Pumping element. The pumping element shall be water-lubricated type and shall consist of a helical-contoured rotor made of heat-treated, chrome-plated, corrosion-resistant steel conforming to ASTM A 582 UNS S41600. The rotor shall be enclosed by an abrasion-resistant, rubber, helical-contoured stator. The pumping element shall require a starting torque value within the limits of 30 inch-pounds to 300 inch-pounds. The torque values shall be for completely wetted stators at 70° F (21° C). The rotor shall be connected to the column shaft by means of a flexible connector cable. The pumping element shall not leak or show permanent deformation when subjected to a hydrostatic pressure of not less than 275 psi.

3.24.2 Head capacity. The Class B pumping assembly shall deliver not less than 50 gpm against a total head of 250 feet while operating at a pump speed of not more than 1,800 rpm.

3.25 Electromagnetic interference. The electromagnetic interference emission characteristics of the pumping assemblies shall conform to MIL-STD-461, Class III A.

3.26 Lubrication. All surfaces requiring lubricants shall be provided with a means for lubricating.

3.26.1 Lubricants. The pumping assemblies shall operate as specified herein when lubricated with military lubricants (see 6.7).

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3.26.2 Lubrication fittings. Lubrication fittings shall conform to MIL-F-3541, Types I, II or III. Fittings shall be located in a protected position and shall be accessible to a hand-operated grease gun conforming to MIL-G-3859 with a 10-inch flexible extension. Accessibility to fittings shall be provided without the removal or adjustment of accessories or parts. Panels and plates equipped with hand-operable, quick-disconnect fasteners may be removed to provide accessibility.

3.26.3 Pressure-release device. A pressure-release device shall be provided where the use of pressure lubricating equipment could damage grease seals or other parts.

3.26.4 Filling, draining, and checking provisions. Enclosures such as gearcases and transmission housings that contain a reservoir of lubricants for the lubrication of the parts enclosed shall be equipped with dip-sticks, checkplugs not less than 1/2-inch pipe size, or sight gages to determine the level of the lubricant. Each enclosure shall be equipped for filling the enclosure with a lubricant and for draining. The drain outlet shall be fitted with a magnetic drainplug, size conforming to MS35844 or MS49006. The drain outlet shall be located so that removal of the plug will result in complete drainage of the lubricant from the enclosure. Drainage shall be to the ground when the pumping assembly is in its normal position. Integral tubes or troughs may be used to convey the lubricant from the drain outlet to the ground.

3.26.5 Grease lubrication. Bearing lubrication, including lubrication of sealed bearings, shall be with greases as follows:

(a) MIL-G-23827 - For instruments, all sealed bearings, and other bearings where operating temperatures do not exceed minus 100° F (minus 73° C) to plus 250° F (plus 121° C).
(b) MIL-G-10924 - For bearings where operating temperatures do not exceed minus 65° F (minus 53.9° C) to plus 225° F (plus 107° C) and in other applications within this temperature range where grease with oxidation-resistant and corrosion-preventive properties is desirable. Not for use on machinery which comes in contact with foods or potable water.

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The pumping assemblies shall be assembled, run-in (broken-in), tested and delivered with these greases. The bearings shall be cleaned before lubrication, because military greases are not always compatible with other greases. A tag shall be attached in a conspicuous place to indicate which military grease has been used.

3.26.6 Oil lubrication. The pumping assemblies shall operate as specified herein when serviced with military oils designated for use in the temperature range specified herein. A tag shall be attached in a conspicuous place to indicate the temperature range, and the lubricant grade used.

3.27 Fungus and moisture resistance. The electrical circuitry, including all components and connections, except as specified below, shall be protected from the effects of fungus growth and moisture by an overall treatment with a varnish conforming to MIL-V-173, Composition I or II, as applicable (see 3.27.1), with 1 percent copper 8-quinolinolate (by weight) based on the nonvolatile content of the varnish.

(a) Components or circuit elements that are inherently fungus and moisture resistant or which are hermetically sealed need not be treated.
(b) Components or circuit elements whose functions will be adversely affected by the varnish coating shall not be treated.

When used, the varnish shall be applied by spray, brush, or a combination of both to give a dry-film thickness of not less than 1 mil to component or element surfaces previously cleaned and prepared so that the surfaces are free from all foreign matter which would interfere with the adherence or function of the varnish.

3.27.1 Composition. Composition II shall be used only in the case when local air pollution regulations governing the application of varnish precludes the use of Composition I. When Composition II is used, the contractor shall provide evidence to the Government that the use of Composition II is required, and shall certify that the Composition II material complies with Rule 102, Southern California Air Pollution Control District.

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3.28 Identification marking. The pumping assemblies shall be identified in accordance with MIL-STD-130. Identification plates shall conform to MIL-P-514, Type I, Style 1, Composition C of Type I, Grade A, Class 1 material.

3.29 Instruction plates. Each pumping assembly shall be equipped with instruction plates or diagrams, including warnings and cautions, describing any special or important procedure to be followed in assembling, operating, or servicing the pumping assembly. Instruction plates shall conform to MIL-P-514, Type III, Composition C, of Type I, Grade A, Class 1 material, dimensions to suit the application. Plates shall be attached by screws, bolts, or rivets in a conspicuous protected location.

3.30 Treatment and painting. The portions of the pumping assembly and its components and parts normally painted shall be cleaned, treated, and painted in accordance with MIL-T-704, Type A, color forest green.

3.31 Stenciling. The gross weight of the power and drive unit shall be stenciled on both sides in such a position as to be readily discernible. The letters shall be Gothic capitals and the numerals shall be of similar appearance. The letters and numerals shall be 2 inches high where space permits. Stenciling shall conform to MIL-T-704.

3.32 Interchangeability. All parts having the same part number shall be functionally and dimensionally interchangeable. Interchangeable parts are defined as two or more parts possessing such functional and physical characteristics as to be equivalent in performance and durability and capable of being exchanged one for the other without alteration of the parts themselves or of adjoining parts except for adjustment and without selection for fit or performance.

3.33 Workmanship.

3.33.1 Metal fabrication. Metal used in fabrication shall be free from kinks and sharp bends. The straightening of material shall be done by methods that will not cause damage to the material. Corners shall be square and true. Flame-cutting, using tips suitable for the thickness of the steel, may be employed instead of shearing and sawing. All bends shall be made with controlled means to insure uniformity of size and shape. Precaution shall be taken to avoid overheating. Heated steel shall be allowed to cool slowly. External surfaces shall be free of burrs, sharp edges and corners, except when sharp edges or corners are required or where they are not detrimental to safety.

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3.33.2 Welding. The surfaces of parts to be welded shall be free from rust, scale, paint, grease, mill scale that can be removed by chipping and wire brushing, and other foreign matter. Welds shall transmit stress without permanent deformation or failure when the parts connected by the welds are subjected to proof and service loading. Parent materials, weld filler metals and fabrication techniques shall be as required to enable the contractor to conform to the examination and test requirements specified in Section 4. Parts to be joined by fillet welds shall be brought into as close contact as possible and in no event shall be separated by more than 0.1875-inch unless appropriate bridging techniques are used. Unless otherwise specified (see 6.2), the welding process used in fabrication of the pumping assemblies shall be at the option of the contractor.

3.33.3 Welders. Before assigning any welder to manual welding work covered by this specification, the contractor shall provide the contracting officer with certification that the welder has passed qualification tests as prescribed by either the following listed codes for the type of welding operations to be performed and that such qualification is effective as defined by the particular code:

AWS D1.1, Structural Welding Code, Section 5, Qualification.
ASME Boiler and Pressure Vessel Code, Section IX, Welding Qualifications.

Contractors who make only horizontal welds need not qualify welders for "all position welding". Subject to approval by the Government, the contractor's standard welder qualification may be substituted in lieu of the above codes provided that the contractor's procedure is equivalent to the above codes. The contractor shall be responsible for determining that automatic welding equipment operators are capable of producing quality welds in accordance with AWS or ASME codes.

3.33.4 Bolted connections. Boltholes shall be accurately formed and shall have the burrs removed. Washers or lockwashers shall be provided where necessary. Matching thread areas securing bolts conforming to SAE J429 or capscrews shall be of sufficient strength to withstand the tensile strength of the bolt. All fasteners shall be correctly torqued and shall have full thread engagement. Bolts shall protrude not more than 2 full threads.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specifications where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards, as applicable.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3).
(b) Initial production inspection (see 4.4).
(c) Quality conformance inspection (see 4.5).
(d) Inspection comparison (see 4.7).
(e) Inspection of packaging (see 4.8).

4.3 First article inspection.

4.3.1 Examination. The pumping assemblies shall be examined as specified in 4.6.1. Presence of one or more defect shall be cause for rejection.

4.3.2 Tests. The pumping assemblies shall be subjected to these tests marked "X" in Column 1 of Table II. Failure of any test shall be cause for rejection.

4.4 Initial production inspection. When specified (see 3.3), one or more initial production pumping assemblies will be selected at random by the Government from the pumping assemblies being produced by production tooling and will be examined as specified in 4.6.1 and subjected to the tests marked "X" in Column 2 of Table II, to determine conformance to the requirements of this specification. The inspection will be

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performed by the Government at a site selected by the Government. Acceptance of an initial production pumping assembly shall not exclude the remaining pumping assemblies from the quality conformance inspection and acceptance provisions specified in Section 4. In addition to any test specified as part of the initial production test, the Government reserves the right to conduct any and all other tests contained in this specification as part of the initial production test, and failure of such additional tests shall have the same effect as failure of those tests specified as initial production tests.

4.4.1 Inspection failure. Failure of an initial production pumping assembly to meet any requirement specified herein during and as a result of the examination and tests specified in 4.4 shall be cause for rejection of the initial production pumping assembly and shall be cause for refusal by the Government to continue acceptance of production pumping assemblies until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government on pumping assemblies previously accepted and produced under the contract. Any deficiencies found as a result of the initial production inspection will be considered prima facie evidence that all pumping assemblies accepted prior to the completion of initial production inspection are similarly deficient unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

4.5 Quality conformance inspection.

4.5.1 Sampling. Sampling for examination and tests shall be in accordance with MIL-STD-105.

4.5.2 Examination. Samples selected in accordance with 4.5.1 shall be examined as specified in 4.6.1. AQL shall be 4.0 percent defective for major defects and 10.0 percent defective for minor defects.

4.5.3 Tests.

4.5.3.1 Individual. Each pumping assembly shall be subjected to the tests marked "X" in Column 3 of Table II. Failure of any test shall be cause for rejection.

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4.5.3.2 Samples. Samples selected in accordance with 4.5.1 shall be subjected to the tests marked "X" in Column 4 of Table II. AQL shall be 2.5 percent defective.

4.6 Inspection procedure.

4.6.1 Examination. The pumping assembly shall be examined as specified herein for the following defects:

Major

101. Dimensions not as specified.
102. Materials not as specified.
103. Components missing or not as specified.
104. Accessories not provided or not as specified.
105. Electromagnetic interference components not identical to the approved model.
106. Workmanship not as specified.

Minor

201. Improper application of materials or coverage of protective coating.
202. Lubricants not as specified or improperly applied.
203. Identification marking missing, incomplete, or illegible.
204. Instruction plate missing or not as specified.
205. Treatment and painting not as specified.
206. Stenciling not as specified.

4.6.2 Tests.

4.6.2.1 Test conditions. Unless otherwise specified in a test, tests shall be performed without shelter and at the climatic conditions existing at the place of test. The pumping assembly shall operate as specified herein without maintenance other than the contractor's recommended normal scheduled maintenance as established by a maintenance schedule prepared and submitted by the contractor prior to test. Prior to the tests, the pumping assembly shall be lubricated with oils and grease specified in 3.26. Oils shall be those designated for use in the ambient temperature at which the tests are being performed.

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4.6.2.2 Test schedule.

TABLE II. Test schedule.

First article	Quality conformance: Initial production	Quality conformance: Individual	Quality conformance: Samples	Test	Test paragraph	Requirement paragraph
1	2	3	4	5	6	7
				Preassembly:		
X	-	-	X	Engine	4.6.2.3.1	3.12
-	X	-	-	Impeller Balance (Class A)	4.6.2.3.2	3.23.4.1.1
-	X	X	-	Pumping Element Torque (Class B)	4.6.2.3.3	3.24.1
-	X	-	-	Environmental	4.6.2.4	3.6
X	X	-	-	Safety	4.6.2.5	3.7
-	X	-	-	Human Factors	4.6.2.6	3.8
X	-	-	-	Noise Level Measurement	4.6.2.7	3.8.1
-	X	-	-	Reliability	4.6.2.8	3.9
-	X	-	-	Maintainability	4.6.2.9	3.10
X	-	-	-	Check Valve Assembly	4.6.2.10	3.14.3
X	-	-	-	Handling Attachments	4.6.2.11	3.14.4
X	-	-	-	Lifting Attachments	4.6.2.12	3.22
X	-	-	X	Performance	4.6.2.13	3.23, 3.24
X	-	-	X	Head Capacity	4.6.2.13.1	3.23.5, 3.24.2
X	-	X	-	Hydrostatic	4.6.2.14	3.23, 3.23.4 3.24, 3.24.1
X	X	-	-	Electromagnetic interference	4.6.2.15	3.25

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4.6.2.3 Preassembly.

4.6.2.3.1 Engine. Prior to assembling the engine to the pumping assembly, the engine shall be tested in accordance with the production engine tests and production control tests of MIL-STD-1410. Nonconformance to the requirements specified herein shall constitute failure of this test.

4.6.2.3.2 Impeller balance (Class A pumping assembly only). Prior to assembly, determine the static balance of the impeller. An out-of-balance condition exceeding 0.50-inch-ounce plus 0.25-inch-ounce per inch of impeller diameter shall constitute failure of this test.

4.6.2.3.3 Pumping element torque (Class B pumping assembly only). Prior to assembly, determine the Class B pump element starting torque. A torque value of less than 30 inch-pounds or more than 300 inch-pounds shall constitute failure of this test.

4.6.2.4 Environmental.

4.6.2.4.1 High temperature. The engine shall be operated under equivalent load required by the applicable pump performance rating for a period of two hours without overheating. Failure of the engine to operate at temperatures specified in 3.6 for two hours or failure of the engine to operate without overheating shall constitute failure of this test.

4.6.2.4.2 Low temperature. The engine shall be operated under equivalent load required by the applicable pump performance rating for a period of 10 minutes. Failure of the engine to start or operate at temperatures specified in 3.6 shall constitute failure of this test.

4.6.2.5 Safety. The pumping assembly shall be evaluated for safety requirements throughout testing as specified in 4.6.2. Nonconformance to 3.7 shall constitute failure of this test.

4.6.2.6 Human factors. The human factors requirements shall be evaluated during testing. Nonconformance to 3.8 shall constitute failure of this test.

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4.6.2.7 Noise level measurement.

4.6.2.7.1 Representative duty position. Measure the noise level at every 45 degree ±5 degree of azimuth, 65 ±1 inches above ground level, at a 5-foot radius from the geometric center of the equipment in accordance with MIL-STD-1474 for steady-state noise. The operating conditions shall be with the pumping assembly installed and immersed in a water source and with the total head equal to the applicable rated head specified in 1.2. Nonconformance to 3.8.1 shall constitute failure of this test.

4.6.2.7.2 Exterior. Measure the noise level in accordance with MIL-STD-1474 for exterior noise. Nonconformance to 3.8.1 shall constitute failure of this test.

4.6.2.7.3 Hazard precautions. Measure the noise level at all control and maintenance positions while operating at the maximum noise condition for those locations. Should the noise level exceed 85 dB(A) at any position, failure to provide the precautions specified in 3.8.1 shall constitute failure of this test.

4.6.2.8 Reliability. The pumping assembly shall be tested in accordance with 4.6.2.13. The pumping assembly shall be continuously operated for a period of 20 hours out of every 24 hours. Normal maintenance shall be conducted during scheduled downtime. Operations and maintenance shall be performed in accordance with the instructions set forth in the manufacturing operation and maintenance schedule. A failure is defined as any malfunction which:

(a) Cannot be corrected within 30 minutes by adjustment, repair, or replacement using only the maintenance tools and repair parts specified to be furnished with the equipment; or
(b) may cause failure to commence operation, cessation of operation, or degradation of performance below specified level; or
(c) may damage the item by continuing operation; or
(d) may cause safety hazard to operating personnel.

The total test time is total unit hours of pump "on" time and is expressed in multiples of specified mean-time-between-failures (MTBF). The test shall continue for a sufficient length of time to reach a decision to "accept" or "reject" in accordance with Figure 1. Nonconformance to 3.9 shall constitute failure of this test.

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4.6.2.9 Maintainability. During testing of the pumping assemblies as specified in 4.6.2.8, adequacy of all the maintenance support elements such as operation and maintenance schedule, lubrication charts, repair parts, and the requirement of common tools as well as special tools shall be evaluated. Any inadequacy in the maintenance support elements, which would result in inability of the pumping assemblies to be maintained as specified, shall constitute failure of this test and shall require revision of the document(s). Errors or inadequacies in the manuals shall not be considered in assessing the maintenance ratio. Nonconformance to 3.10 shall constitute failure of this test.

4.6.2.10 Check valve assembly. The check valve assembly shall be subjected to a pressure of not less than 150 psi for a period of not less than 2 minutes. Exterior leakage, back flow through the valve, or permanent deformation shall constitute failure of this test.

4.6.2.11 Handling attachments. Subject the handling attachments on the discharge head to a weight equal to the load capability specified in 3.14.4. Cracking, breaking, or permanent deformation of the handling attachments shall constitute failure of this test.

4.6.2.12 Lifting attachments. Subject the lifting attachments on the power and drive unit to the loads specified in 3.22. Cracking, breaking, or permanent deformation of the lifting attachments shall constitute failure of this test.

4.6.2.13 Performance. The pumping assembly shall be subjected to performance tests while pumping water at the operating conditions specified herein. Tests of the Class B pump shall be performed in accordance with the test code of the Standards of the Hydraulic Institute for Rotary Pumps (Single Screw Type). Tests of the Class A pump shall be conducted in accordance with the applicable provisions of the test code of AWWA E101. Curves shall be plotted from data obtained by the tests to show gpm versus total head. The brake horsepower shall be plotted on the same graph. All test data shall be corrected to standard sea-level atmospheric conditions of 29.92 inches of mercury and to the specific weight of water (62.3 pounds per cubic foot) at temperature of 68° F.

4.6.2.13.1 Head capacity. Operate the pumping assembly to determine the head capacity of the pumps as specified herein. The operating time shall be that required to obtain the performance characteristics

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of the pumps. The pumping element shall be at least 10 feet below the level of the skid base.

4.6.2.13.1.1 Class A, turbine. Operate the pumping assembly at a speed of not more than 3,600 rpm. Capacities of less than specified in 3.23.5 at 200-foot total head for Size 1 pump, 350-foot total head for Size 2 pump, or 250-foot total head for Size 3 pump shall constitute failure of this test.

4.6.2.13.1.2 Class B, helical rotor. Operate the pumping assembly at a pumping element speed of not more than 1,800 rpm. During the head-capacity test, set the pressure-relief valve specified in 3.23 at 35 psi, 60 psi, and 100 psi, respectively. Head capacity of less than that specified in 3.24.2 or failure of the pressure relief valve to release at the specified pressures shall constitute failure of this test.

4.6.2.14 Hydrostatic. Subject the discharge head assembly and pumping element to the hydrostatic pressure specified in 3.23 and 3.23.4 for Class A, or 3.24 and 3.24.1 for Class B, as applicable for not less than 5 minutes. Leakage, failure, or permanent deformation of any component of the discharge head assembly or pumping element shall constitute failure of this test.

4.6.2.15 Electromagnetic interference. The pumping assembly shall be tested to determine conformance to 3.25. The contractor shall furnish the contracting officer the report of test required by MIL-STD-461. Disapproval of the report shall constitute failure of this test.

4.7 Inspection comparison. When initial production inspection is specified, the Government may select pumping assemblies at any time during the contract production period and subject these pumping assemblies to the examination specified in 4.6.1 and the tests marked "X" in Column 2 of Table II to determine conformance to the requirements of this specification. The inspection will be performed by the Government, at a site selected by the Government, on units selected at random from those which have been accepted by the Government and will not include the previously inspected first article and initial production pumping assemblies. In addition to any test specified as part of the inspection comparison, the Government reserves the right to conduct any and all other tests contained in this specification as part of the inspection comparison and failure of such additional tests shall have the same effect as failure of those tests specified as inspection comparison.

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4.7.1 Inspection failure. Failure of an inspection comparison pumping assembly to meet any requirement specified herein during and as a result of the examination and tests specified in 4.7 shall be cause for rejection of the inspection comparison pumping assemblies and shall be cause for refusal by the Government to continue acceptance of production pumping assemblies until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government on pumping assemblies previously accepted and produced under the contract. Any deficiencies found as a result of the inspection comparison will be considered prima facie evidence that all pumping assemblies accepted prior to the completion of inspection comparison are similarly deficient unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

4.8 Inspection of packaging. The preservation packing, and marking for Level A and Level B shall be examined and tested to determine compliance with the quality assurance provisions of MIL-P-710. Commercial preservation, packing and marking shall be examined for conformance to MIL-STD-1188.

5. PACKAGING

5.1 Preservation, packing and marking. Preservation, packing and marking shall be Level A, Level B or Commercial as specified (see 6.2).

5.1.1 Level A and Level B. Level A and Level B preservation, packing and marking shall be in accordance with MIL-P-710.

5.1.2 Commercial. Commercial preservation, packing and marking shall be in accordance with MIL-STD-1188.

6. NOTES

6.1 Intended use. The pumping assemblies are intended for use in semi-permanent and permanent deep-well, water-supply systems.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title number, and date of this specification.
(b) Class and size of pump required (see 1.2).

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(c) When a first article inspection is not required (see 3.2).
(d) Time frame required for submission of first article and number of pumping assemblies to be furnished (see 3.2).
(e) When the Government will conduct any or all of the first article inspections. When the Government will conduct some but not all of the first article inspections, the contracting officer should specify which inspections will be conducted by the Government and which inspections shall be conducted by the contractor (see 3.2).
(f) When initial production inspection is required and number of pumping assemblies to be furnished (see 3.3).
(g) Whether liquid or air-cooled engine is required (see 3.12).
(h) When cold starting aid is required (see 3.12.7).
(i) When welding process shall not be at the contractor's option (see 3.33.2).
(j) Degree of preservation and degree of packing required (see 5.1).

6.3 First article sample. Any changes or deviations of production pumping assemblies from the approved first article during production will be subject to the approval of the contracting officer. Approval of the first article will not relieve the contractor of his obligation to furnish pumping assemblies conforming to this specification.

6.4 Data requirements. The contracting officer should include requirements for such data as technical publications, instructional materials, illustrated parts lists, and contractor's maintenance and operation manual to be furnished with each pumping assembly.

6.5 Provisioning. The contracting officer should include provisioning requirements for repair parts and maintenance tools as necessary (including any special tools), and instructions on shipment of pumping assemblies.

6.6 Noise limits. When the noise level of the pumping assembly exceeds 85 dB(A), the appropriate discussion of the noise hazard in accordance with MIL-STD-1474 (MANUALS Paragraph) shall include the requirement for hearing protection, the noise level of the pumping assembly, the distance from the pumping assembly at which 85 dB(A) will always be met, and what operational situation or system configuration will reduce the level to 85 dB(A).

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6.7 Lubricants. The contracting officer should furnish a list of military lubricants applicable to the pumping assemblies covered by this specification as contained in the Federal Supply Catalog, Department of Defense Section, Identification List C9100-IL, for FSC Group 91.

6.8 Recycled material. It is encouraged that recycled material be used, when practical, as long as it meets the requirements of this specification (see 3.4).

Custodians:
Army - ME
Navy - YD

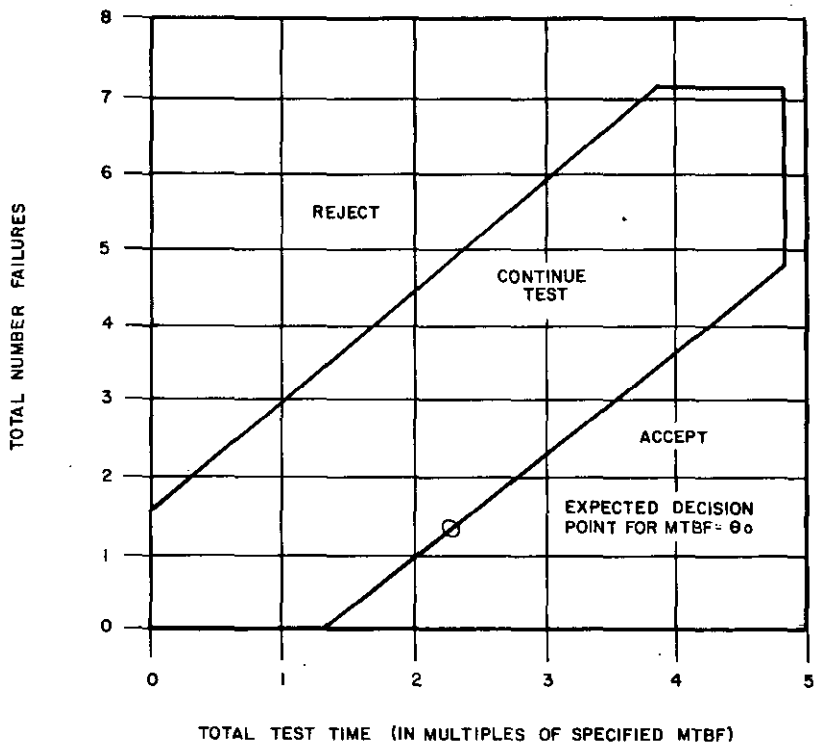
Preparing activity:
Army - ME

Review activity:
DLA - CS

Project 4320-0282

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DECISION RISKS 20%
DISCRIMINATION RATIO 2.0:1



NO. OF FAILURES	TOTAL TEST TIME * REJECT (EQUAL OR LESS)	ACCEPT EQUAL OR MORE
0	N/A	1.40
1	N/A	2.09
2	.35	2.79
3	1.04	3.48
4	1.73	4.17
5	2.43	4.87
6	3.12	4.87
7	3.81	4.87
8	4.87	N/A

*TOTAL TEST TIME IS TOTAL UNIT HOURS OF "EQUIPMENT ON" TIME AND IS EXPRESSED IN MULTIPLES OF THE SPECIFIED MTBF.

FIGURE 1. Accept-reject criteria.

X-3661

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DOCUMENT IDENTIFIER *(Number)* AND TITLE MIL-P-52029D Pumping Assemblies, Deep-Well, GED: Helical-Rotor and Turbine Types

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1. ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDED WORDING CHANGE

C. REASON FOR RECOMMENDED CHANGE(S)

2. REMARKS

SUBMITTED BY *(Printed or typed name and address — Optional)*

TELEPHONE NO.

DATE

DD FORM 1426 1 OCT 76

EDITION OF 1 JAN 72 WILL BE USED UNTIL EXHAUSTED.

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